FP Phase 2 EDA, baseline pipeline, Scalability, Efficiency, Distributed/parallel Training, and Scoring Pipeline

- Due Nov 24, 2025 by 3:59pm
- Points 100
- Submitting a file upload
- Available Oct 20, 2025 at 3:30pm - Dec 1, 2025 at 3:59pm

This assignment was locked Dec 1, 2025 at 3:59pm.

Phase II - EDA, baseline pipeline on all available data, Scalability, Efficiency, Distributed/parallel Training, Scoring Pipeline, Feature engineering and hyperparameter tuning

Note: By design, every aspect of the final project is up to the interpretation of your team. Don't think of the objectives as a checklist! The only things required for Phase 02 are to have a working baseline on 1-year data, and the EDA on 1-year data. Having said that, please aim to have 5 years ready!

Objectives

In this phase of the project, please perform the following tasks:

- EDA on all tables (raw data) - This EDA can either be simple (data size, missing value analysis, duplicate analysis), or more intricate. However, given that you will use OTPW as your main data source, you can focus on that only. Now: if you are planning to create your own join (See *Extra Credit*), then you have to do the intricate EDA.
    - Airline On-Time Performance Data (Flights)
        - **https://www.transtats.bts.gov/Tables.asp?QO_VQ=EFD&QO_anzr=Nv4yv0r%FDb0-gvzr%FDcr4s14zn0pr%FDQn6n&QO_fu146_anzr=b0-gvzr** (https://www.transtats.bts.gov/Tables.asp?QO_VQ=EFD&QO_anzr=Nv4yv0r%FDb0-gvzr%FDcr4s14zn0pr%FDQn6n&QO_fu146_anzr=b0-gvzr)
    - Weather
    - Weather Stations
    - Airport codes data
        - **https://datahub.io/core/airport-codes** (https://datahub.io/core/airport-codes)
- **The ATP (Airline On-Time Performance Data) flight data has been pre-joined with the Weather data to yield the OTPW (the joined ATP and Weather) dataset.**

- **Please do EDA on OTPW (the joined ATP and Weather) based on 3 months and 12 months*:***
  - Three months of OTPW data. (Expected for Phase 2)
  - One first-year of OTPW. (Expected for Phase 2)
  - The entire OTPW dataset. (Good to have for Phase 2, Expected for Phase 3)
- Address missing data
  - Quantity
  - Discuss why there is so much missing data for various attributes
- Address non-numerical features
- List out raw features, derived features that you plan to implement/use
- Do you need any dimensionality reduction? (e.g., LASSO regularization, forward/backward selection, PCA, etc..)
- Specify the feature transformations for the pipeline and justify these features given the target (e.g., hashing trick, tf-idf, stopword removal, lemmatization, tokenization, etc..)
- Other feature engineering efforts, i.e., interaction terms, Breiman's method, etc.
- Review the following material regarding developing machine learning pipelines in Spark:
  - **https://pages.databricks.com/rs/094-YMS-629/images/02-Delta%20Lake%20Workshop%20-%20Including%20ML.html** (https://pages.databricks.com/rs/094-YMS-629/images/02-Delta%20Lake%20Workshop%20-%20Including%20ML.html)
  - **https://spark.apache.org/docs/latest/ml-tuning.html** (https://spark.apache.org/docs/latest/ml-tuning.html)
- Create machine learning baseline pipelines using logistic/linear regression and ensemble models and do experiments on the following:
  - Three-month OTPW dataset (this is **optional** but is a great stepping stone towards the one-year dataset)
  - One-year OTPW dataset
  - Log and report on experimental results (with a primary focus on the one-year OTPW dataset)
    - For the one-year OTPW dataset, please use the last quarter of the available one-year dataset as a blind test set that is never consulted during training.
    - Report evaluation metrics in terms of cross-fold validation over the training set (first three-quarters of the one-year of OTPW dataset)
    - Report evaluation metrics in terms of the blind test set.
  - Discuss experimental results and main findings.
  - **Hint**: **cross-validation in Time Series** is very different from regular cross-validation. Please review the following to get more background on CV for time-based data.
    - **Cross-validation in Time Series data (very different from regular cross-validation)**
      - The method that can be used for cross-validating the time-series model is cross-validation on a rolling basis. Start with a small subset of data for training purposes, forecast for the later data points, and then check the accuracy of the forecasted data points. The same forecasted data points are included as part of the next training dataset, and subsequent data points are forecasted.

- For more information, see:
  - [https://hub.packtpub.com/cross-validation-strategies-for-time-series-forecasting-tutorial/ (https://hub.packtpub.com/cross-validation-strategies-for-time-series-forecasting-tutorial/)](https://hub.packtpub.com/cross-validation-strategies-for-time-series-forecasting-tutorial/)
  - [https://medium.com/@soumyachess1496/cross-validation-in-time-series-566ae4981ce4#:~:text=Cross%20Validation%20 (https://medium.com/@soumyachess1496/cross-validation-in-time-series-566ae4981ce4#:~:text=Cross%20Validation%20)](https://medium.com/@soumyachess1496/cross-validation-in-time-series-566ae4981ce4#:~:text=Cross%20Validation%20)

- Create new features that are highly predictive:
  - at least one time-based feature, e.g., recency, frequency, monetary, (RFM)
- Fine-tune your baseline pipeline using a grid search
  - Is there a difference in performance? Is it related to features? Is it related to noise? What is impacting the model performance?

DELIVERABLES:

1. **Your project notebook for this phase. Absolutely no code in the report notebook unless you plan to show or highlight an important feat that your team accomplished. Your code notebooks should be referenced at the end of the report.**
2. **Updated Credit assignment plan (at the top of your notebook: list of SMART Goals with the person responsible for each)**
3. **Please highlight the phase leader for this phase.**
4. **In-class Presentation**

- **EXTRA CREDIT:** Join your own flights and weather tables (full join of all the data) and generate the dataset that will be used for training and evaluation.
  - Joins take 2-3 hours with 10 nodes;
    - Join stations data with flights data
    - Join weather data with flights + Stations data
  - EDA on joined dataset that will be used for training and evaluation - **Up to 15 points of Extra Credit (this is a though task, only try if you feel that you will complete it, priority should be the rest of the project) The Extra Credit can be claimed on any phase (2 or 3)**

Submission Requirements and format

You will submit the following items:

1. DataBricks notebook in .html format
   - Please make sure the links/images are visible to others by viewing your file in incognito mode on your browser
   - Your submission will be penalized if you do NOT meet these requirements

Final Project: Phase 2 (1)

Criteria	Ratings			Pts
In class presentation In-class presentation * In-class explanation, and discussion of your results (as part of the final exam period)	**20 to >10.0 pts** **Full Marks! Awesome Presentation!**	**10 to >0.0 pts** **Flow of in-class presentation is not perfect** Please make sure your presentation has a title slide (with the project name, Group Number, team member names, and photos) has an outline slide with good descriptive section headings. -- your presentation has a logical business manner: You need to cover the following in your presentation: -- Team member names, Group Number, photos -- Project description -- Some summary visual EDA -- Overview of Modeling Pipelines explored -- Results and discussion of results -- Conclusion and next steps	**0 pts** **No Marks: your presentation does not meet my expectations for young data scientist.** Your presentation does not meet my expectations for young a data scientist team. Please review what was expected as a submission for this phase of the project. And reconcile where you fell short.	20 pts
Team and project meta information. Highlight the Phase leader who is the contact person for this phase.	**3 pts** **Complete team and project information.**	**1.5 pts** **Team and project meta information is incomplete and or not formatted correctly** Team and project meta information is incomplete and or not formatted correctly To improve the quality of this section try the following: Please make a section and subsection for each of the following. - Team	**0 pts** **No Marks: this does not meet my expectations for young data scientist.** This section does not meet my expectations for young a data scientist. Communication is key in data science and not getting this section right is not acceptable. Please review what was expected as a	3 pts

Criteria	Ratings			Pts
		information -- Phase Leader (make sure this is visible in the top of the notebook) -- Project Title -- Group number -- Names, emails -- Team member photos -- format the photos to be the same size and label each photo with the person's name -- Or merge all photos to make a montage	submission for this phase of the project. And reconcile where you fell short.	
Project Abstract Abstract: You now have some results! So your abstract should be an evolved version of your Phase 1's abstract (plus any feedback that you got!). For Phase 2: 1.- Repeat the same components of Phase 1: Business problem, data used and sources, etc. 2.- Explain your baseline: why did you select it and what's the results of your metric in the baseline. 3.- What are the most important Features that you are using/planning to use. 4.- Next steps. Always be concise. This abstract should	**10 pts** **Full Marks! Excellent abstract!**	**5 pts** **Abstract is incomplete and or is not formatted correctly** Abstract is incomplete and or is not formatted correctly To improve the quality of this section. Then make sure to describe what you focused on and accomplished in this phase. Have a look at the expectations with regard to what a good abstract should include and how it should be presented.	**0 pts** **No Marks: your abstract does not meet my expectations for young data scientist.** Your does not meet my expectations for young a data scientist. Communication is key in data science and not providing a good abstract is not acceptable. Please review what is expected as an abstract for this phase of the project and reconcile where you fell short.	10 pts

Criteria	Ratings			Pts
have between 2 or 3 paragraphs top.				
Project Description (data and tasks) Expectations here are to provide the following in sections and subsections: Description of the data and task at hand --Data description --Task to be tackled -- Provide diagrams to aid understanding the workflow	**10 pts** **Full Marks! Excellent Project description!**	**5 pts** **Project description is incomplete and or is not formatted correctly** Project description is incomplete and or is not formatted correctly To improve the quality of this section: Provide the following in sections and subsections: Description of the data and task at hand -- Data description -- Task to be tackled -- Provide diagrams to aid understanding the workflow EDA is incomplete and or is not formatted correctly To improve the quality of this section: Provide the following in sections and subsections: -- A data dictionary of the raw features (test description; data type: numerical, list, etc.) -- Dataset size (rows columns, train, test, validation) -- Summary statistics -- Correlation analysis -- Other useful text-based analysis (as opposed to graphic-based)	**0 pts** **No Marks: your project description does not meet my expectations for young data scientist.** Your project description does not meet my expectations for a young data scientist. Communication is key in data science and not providing a good project description is not acceptable. Please review what is expected as a project description for this phase of the project and reconcile where you fell short.	10 pts
EDA Expectations here are to provide the	**10 pts** **Excellent EDA.**	**5 pts** **EDA is incomplete and or is not formatted correctly**	**0 pts** **No Marks: your EDA does not meet my**	10 pts

Criteria	Ratings			Pts
following in sections and subsections: EDA -- Summary description of each table in English. -- A data dictionary of the raw features (test description; data type: numerical, list, etc.) -- Dataset size (rows columns, train, test, validation) -- Summary statistics -- Correlation analysis -- Other useful text-based analysis (as opposed to graphic-based) Here are some ideas of what it should have, you DON'T NEED to do all of them! -- A visualization of each of the input and target features (looking at the distribution, and the central tendencies as captured by the mean, median etc.) -- A visualization of the correlation analysis -- pair-based visualization of the input and output features		EDA is incomplete and or is not formatted correctly To improve the quality of this section: Provide the following in sections and subsections: -- A data dictionary of the raw features (test description; data type: numerical, list, etc.) -- Dataset size (rows columns, train, test, validation) -- Summary statistics -- Correlation analysis -- Other useful text-based analysis (as opposed to graphic-based)	**expectations for young data scientist.** Your EDA does not meet my expectations for a young data scientist. Communication is key in data science and not providing a good EDA is not acceptable. Please review what is expected as an EDA for this phase of the project and reconcile where you fell short.	

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Overall, make sure your graphics are: well sized, readable; have titles, have xlabels with units, have ylabels with units; bonus points for graphics with annotations! -- a graphic summary of the missing value analysis -- other novel visualizations (e.g., geo-plot)				
Modeling Pipelines Expectations here are to provide the following in sections and subsections: Modeling Pipelines -- A visualization of the modeling pipeline (s) and sub pipelines if necessary -- Families of input features and count per family -- Number of input features -- Loss function used (data loss and regularization parts) in latex -- Number of	**20 pts** **Great pipeline code, comments and description for the new features.** Great pipeline code, comments, and description for the new features (each with its own sub-section):	**10 pts** **Modeling pipelines are incomplete and or running properly** Modeling pipelines are incomplete and or running properly To improve the quality of this section: Make sure to provide the following in sections and subsections: -- Provide a block diagram of your modeling pipeline -- Make sure your pipelines are documented -- Verify your pipelines are complete and execute properly -- Your experiments are not documented properly (the are missing	**0 pts** **No Marks: your Modeling Pipelines section does not meet my expectations for young data scientist.** Your Modeling Pipelines section does not meet my expectations for a young data scientist. Experimentation and implementation are core skills in being a data scientists. In your submission both of these skills are lacking. Please review what is expected as a Modeling Pipelines for this phase of the project and reconcile where you fell short.	20 pts

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experiments conducted -- Describe cluster size, and how many minutes it takes to build each model type. -- Experiment table with the following details per experiment: ----- Baseline experiment ---- The families of input features used ----- For train/valid/test record the following in a Pandas DataFrame: ---- List Metrics used along with: their Latex equations		accurate descriptions, performance metrics)		
Results and discussion of results Expectations here are to provide the following in sections and subsections: Please report the results of training (first three-quarters of the one-year dataset), and blind testing (the last quarter of the available one-year dataset). The aim of the discussion section is	**10 pts** **Great results and discussion of results.**	**5 pts** **Results and discussion are incomplete and or not adequately formatted** Results and discussion are incomplete and or not adequately formatted To improve the quality of this section: Make sure to provide the following in sections and subsections: -- Your experiments are not properly enumerated/tabulated and discussed (they are missing accurate descriptions, performance metrics) --	**0 pts** **No Marks: your results and discussion does not meet my expectations for young data scientist.** Your results and discussion do not meet expectations for a young data scientist. If results are not reported with respect to the full dataset, this will results in a ZERO grade for this aspect of your Phase 2 submission: I.e., please report the results of training , and blind testing. Experimentation	10 pts

Criteria	Ratings			Pts
to review experiments conducted and draw conclusions about key pipelines (supported by experimental results).. Often, this part is the most important, simply because it lets the researcher take a step back and give a broader look at the experiment. Do not discuss any outcomes not presented in the results part (not supported by experiments).		Do not discuss results not substantiated in your experimental section above in the modeling pipelines	presentation and discussion are core skills in being a data scientist. In your submission, both of these skills are lacking. Please review what is expected as a result and discussion for this project phase and reconcile where you fell short.	
Conclusion Expectations here are to address the following following in your conclusion (in about 150 words) in a main section by itself: -- Restate your project focus explain why it's important. Make sure that this part of the conclusion is concise and clear. -- Restate your hypothesis (e.g., ML pipelines with custom features can accurately forecast	**10 pts** **Full Marks! Excellent Conclusion!**	**5 pts** **Conclusion are incomplete and or does not flow well.** Conclusion is incomplete and or does not flow well. To improve the quality of this section: -- Make sure your conclusion is about 150 long or less. -- That you address all the expected points in a conclusion adequately.	**0 pts** **No Marks: your conclusion does not meet my expectations for young data scientist.** Your conclusion does not meet my expectations for a young data scientist. Communicating a conclusion is a core skill in being a data scientists. Please review what is expected as content, style, and format for a conclusion for this phase of the project and reconcile where you fell short.	10 pts

Criteria	Ratings			Pts
.....) -- Summarize the main points of your project: Remind your readers of your key contributions. -- Discuss the significance of your results -- Discuss the future of your project and closing thoughts.				
Updated Credit assignment plan Credit assignment plan updates (who does/did what and when) in Table format No Credit assignment plan means ZERO points Credit assignment plan not in Table format means ZERO points No start and end dates and budgeted hours of effort means an incomplete plan. May result in zero points.	**7 pts** **Full Marks**	**0 pts** **No Marks**		7 pts
Joining the data [Extra Credit]	**15 pts** **Great Visual.**	**7.5 pts** **Joins are incomplete and or is not**	**0 pts** **No Marks: your table join does not meet**	15 pts

Criteria	Ratings			Pts
Expectations here are to provide the following in sections and subsections: -- steps in your joins in words, code, and data sizes for all the data -- Joins need to be reported and performed on all the data (not just 3/6/12 months). -- join statistics (table sizes, time to run joins, and cluster sizes)		**formatted correctly** Joins were not completed on the full data or Joins were not described adequately No join statistics (table sizes, time to run joins, cluster sizes were not provided)	**expectations for a young data scientist.** Your joins do not meet expectations for a young data scientist.	
Total Points: 115				